

DIFFERENTIAL EQUATIONS

LECTURE NOTES

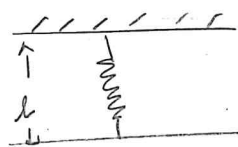
WEEK X

MECHANICAL VIBRATIONS

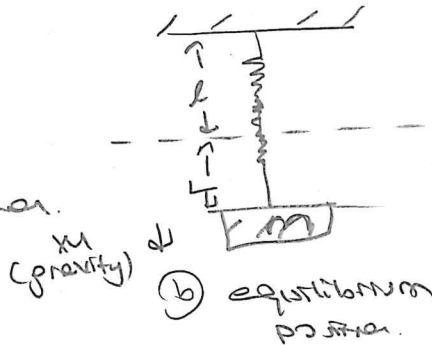
The equation of the form

$$m\ddot{u} + \gamma\dot{u} + ku = F(t)$$

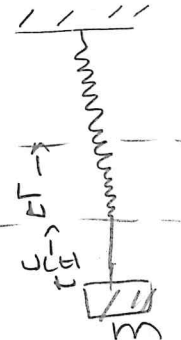
governs the motion of a vibrating mass at the end of a vertical spring, where m, γ, k are positive constants and F is a given function (external force) and $u(t)$ is the displacement of a mass at time t .



(a) natural position.



(b) equilibrium position.



(c) mass in motion.

$u(t)$: displacement of mass from its equilibrium position at time t .

→ To find $u(t)$, the position of the mass m at time t , we consider the forces acting on the mass.

(1) The gravitational force $\boxed{xu = mg}$ (1) is the weight of mass m .
 weight $\downarrow$ gravity acceleration

• $\left[g = 9.8 \text{ m/s}^2 = 32 \text{ ft/s}^2 \right]$ $\uparrow mg (2.5 \text{ cm})$
 $1 \text{ m} = \frac{1}{12} \text{ ft}$ $\rightarrow$ + foot (200)
 is in the downward direction, which we take as positive

② The restoring force (Spring force F_s) of the spring. The spring is stretched ($u+L$) units beyond its natural length

Hence, the magnitude of F_s is $k(u+L)$, where k is the spring constant by Hooke's law

Since the spring pulls upward (negative direction)

$$\boxed{F_s = -k(u+L)} \quad (2)$$

When $u=0$, that is, the system is at equilibrium

F_s and W balance each other. Thus,

$$mg - kL = 0 \Rightarrow mg = kL \Rightarrow k = \frac{mg}{L} = \frac{W}{L}$$

③ Damping force, F_d is the damping which the medium exerts on the mass m .
at all time or damping
here and there

Damping force $\propto$ magnitude of the velocity of the mass but opposite direction

$$F_d \propto -u'(t) = \frac{du}{dt}$$

$$\boxed{F_d = -\gamma u'(t)} \quad (3) \text{ } \gamma \text{ damping constant.}$$

④ The force F is the external force applied to mass m .
 The force may be directed upward or downward

Then, from Newton's Second law of motion

$$\begin{array}{c} f \\ \downarrow \\ \text{net total force} \end{array} = ma \rightarrow \text{acceleration} \Rightarrow \boxed{f = m u''} \quad (4)$$

$$m\ddot{u} = u + F_s + F_d + F(t) = mg - k(u+L) - \gamma \dot{u}' + F(t)$$

$$= \underbrace{mg - kL - kL}_{=0 \text{ equilibrium}} - \gamma \dot{u}' + F(t)$$

$$m\ddot{u} + \gamma \dot{u}' + ku = F(t)$$

$m, \gamma, k > 0.$

" 2nd order linear DE with " constant coefficient

Definition 1 when $\gamma = 0$, the system is called undamped otherwise damped

2 when $F(t) = 0$, the motion is called free, otherwise it is called forced motion

SPECIAL CASES

1 Undamped free vibration

$$\gamma = 0 \quad F(t) = 0$$

$$m\ddot{u} + ku = 0 \Rightarrow \ddot{u} + \frac{k}{m}u = 0$$

$$\lambda^2 + \frac{k}{m} = 0 \Rightarrow \lambda^2 = -\frac{k}{m}$$

$$\lambda_{1,2} = \mp i \sqrt{\frac{k}{m}} = \mp i \omega_0$$

$\omega_0 = \sqrt{\frac{k}{m}}$: natural frequency of the motion of mass.

Then, $u(t) = A \cos(\omega_0 t) + B \sin(\omega_0 t)$

which is a periodic motion with period $T = \frac{2\pi}{\omega_0}$

We can express $u(t)$ in the form

$u(t) = R \sin(\omega_0 t + \phi)$ OR $u(t) = R \cos(\omega_0 t - \phi)$

Let $A = R \sin \phi$ $B = R \cos \phi$

$u(t) = R (\sin \phi \cos(\omega_0 t) + \cos \phi \sin(\omega_0 t))$

$R = \sqrt{A^2 + B^2}$ (extreme displacement) $\rightarrow$ amplitude of motion max displacement of mass from equilibrium

$\tan \phi = \frac{A}{B}$ (phase angle)

$u(t) = R \cos \phi \cos(\omega_0 t) + R \sin \phi \sin(\omega_0 t) = R \cos(\omega_0 t - \phi)$

$R = \sqrt{A^2 + B^2}$ (max displacement)

Ex A mass weighting 4 lb ($1 \text{ lb} \approx 0.5 \text{ kg}$) stretches a spring 2 in . Suppose that the mass is displaced an additional 6 in in the positive direction and then released. The mass in the medium that exerts a viscous resistance of 6 lb when the mass has a velocity 3 ft/s . Find I.V.P.

$$W = 4 \text{ lb}$$

$$W = mg$$

$$L = 2 \text{ in} = 2 \cdot \frac{1}{12} = \frac{1}{6} \text{ ft}$$

$$m = \frac{W}{g} = \frac{4}{32} = \frac{1}{8}$$

$$W = kL \Rightarrow k = \frac{W}{L} = \frac{4}{1/6} = 24$$

$$F_d = 6 \text{ lb} \quad v'(t) = 3 \text{ ft/s} \quad \gamma = \frac{F_d}{v'(t)} = \frac{6}{3} = 2$$

no external force $F(t) = 0$

$$m v'' + \gamma v' + k v = F(t)$$

$$\frac{1}{8} v'' + 2 v' + 24 v = 0$$

$$\Rightarrow v'' + 16 v' + 192 v = 0$$

I.V.P. $t=0$: $v(0) = 6 \text{ in} = 6 \cdot \frac{1}{12} = \frac{1}{2} \text{ ft}$ (displaced in positive direction)
 $v'(0) = 0$ (mass is released)
 $v(0) = \frac{1}{2} \quad v'(0) = 0$
 $\lambda^2 + 16\lambda + 192 = 0$
 $\lambda_{1,2} = ?$

Positive way $\downarrow$ because of motion.

Ex 7. A mass weighting 10 lb stretches a spring 2 in . If the mass is displaced an additional 2 in and is then set in motion with an initial $\uparrow$ upward velocity of $\frac{1}{6}\text{ ft/s}$. Determine the position of mass at any later time. Find period, amplitude and phase of the motion.

$m = 10\text{ lb}$ $L = 2\text{ in} = \frac{1}{6}\text{ ft}$ $u(0) = 2\text{ in} = \frac{1}{6}\text{ ft}$ $u(t) = ?$
 $u'(0) = -\frac{1}{6}\text{ ft/s}$ (Upward negative)

No damping $\gamma = 0$
 No external force $F(t) = 0$

} Underdamped, free motion

$m = mg \Rightarrow n = \frac{m}{g} = \frac{10}{32}\text{ lb/ft s}^2$ $\omega = kL \Rightarrow k = \frac{\omega}{L} = \frac{10}{1/6} = 60$

$\frac{10}{32} u'' + 60u = 0$ $u(0) = \frac{1}{6}, u'(0) = -1$

$u'' + 192u = 0 \Rightarrow \lambda^2 + 192 = 0 \Rightarrow \lambda^2 = -192 \Rightarrow \lambda_{1,2} = \pm 8\sqrt{3}i$

$u(t) = A \cos(8\sqrt{3}t) + B \sin(8\sqrt{3}t)$

$u(0) = \frac{1}{6} = A$

$-1 = u'(0) = (-A \sin(8\sqrt{3}t) \cdot 8\sqrt{3} + 8\sqrt{3} B \cos(8\sqrt{3}t)) \Big|_{t=0}$

$B = -\frac{1}{8\sqrt{3}}$

$u(t) = \frac{1}{6} \cos(8\sqrt{3}t) - \frac{1}{8\sqrt{3}} \sin(8\sqrt{3}t)$

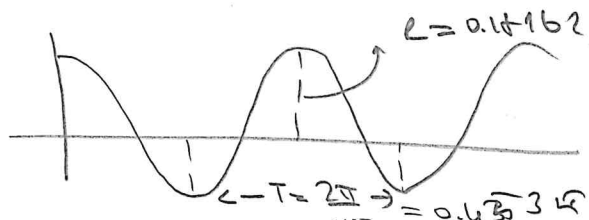
Amplitude $= R = \sqrt{A^2 + B^2} = \sqrt{\left(\frac{1}{6}\right)^2 + \left(-\frac{1}{8\sqrt{3}}\right)^2} \approx 0.1162$

$A = R \cos \phi \quad B = R \sin \phi \Rightarrow \tan \phi = \frac{B}{A} = \frac{-\frac{1}{8\sqrt{3}}}{\frac{1}{6}} = -\frac{3}{4\sqrt{3}} = -\frac{\sqrt{3}}{4}$

$\phi = \arctan\left(-\frac{\sqrt{3}}{4}\right)$

$u(t) = 0.1162 \cos(\omega_0 t - \phi) = 0.1162 \cos(\sqrt{192} t - \phi)$

$\omega_0 = \sqrt{\frac{k}{m}} = \sqrt{\frac{60}{10/32}} = \sqrt{192}$



(7)

② Damped Free Vibration

$$\gamma \neq 0, F(t) = 0$$

$$m u'' + \gamma u' + k u = 0$$

$$m \lambda^2 + \gamma \lambda + k = 0$$

$$\lambda_{1,2} = \frac{-\gamma \pm \sqrt{\gamma^2 - 4km}}{2m}$$

We have 3 cases

① If $\Delta = \gamma^2 - 4km > 0 \Rightarrow$ we have two distinct real roots λ_1, λ_2 .

$$u(t) = A e^{\lambda_1 t} + B e^{\lambda_2 t} \Rightarrow \Delta > 0$$
 the system is "overdamped"

$$\lambda_1 = \frac{-\gamma + \sqrt{\gamma^2 - 4km}}{2m}$$

$$\lambda_2 = \frac{-\gamma - \sqrt{\gamma^2 - 4km}}{2m}$$

$0 < \gamma^2 - 4km < \gamma^2 \Rightarrow$ both λ_1 and λ_2 are negative

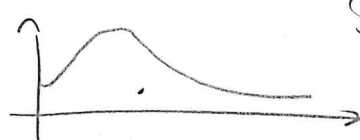
So as $t \rightarrow \infty$ $u(t) \rightarrow 0$

Behavior

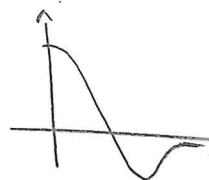
①



②



③



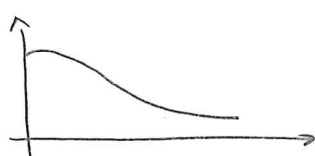
② If $\Delta = \gamma^2 - 4km = 0 \Rightarrow$ critically damped motion

$$\lambda_{1,2} = -\frac{\gamma}{2m}$$

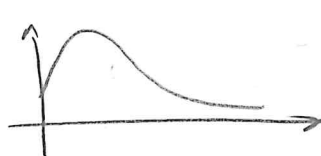
$$u(t) = (A + Bt) e^{-\gamma/2m t} \rightarrow 0 \text{ as } t \rightarrow \infty$$

Behavior

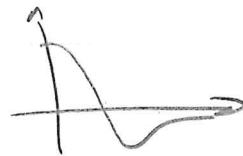
①



②



③



$$u'(t) = \left(B - \frac{\gamma}{2m} A - \frac{\gamma}{2m} Bt \right) e^{-\gamma/2m t} = 0$$

$$\text{C.I.P.S: } B - \frac{\gamma A}{2m} - \frac{\gamma}{2m} Bt = 0$$

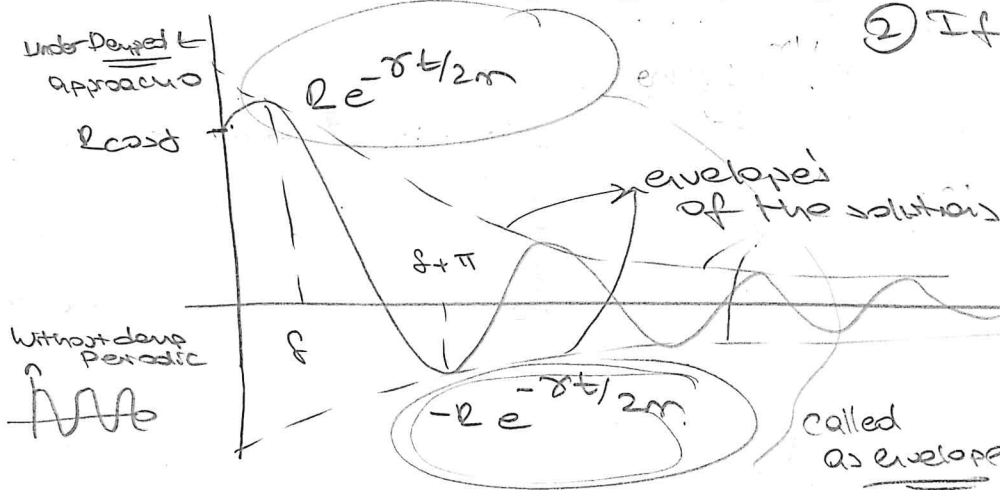
3 $\Delta = \gamma^2 - 4km < 0 \Rightarrow$ "underdamped motion"

$$\lambda_{1,2} = \frac{-\gamma \pm \sqrt{\gamma^2 - 4km}}{2m} = \frac{-\gamma \pm \sqrt{4km - \gamma^2} i}{2m}$$

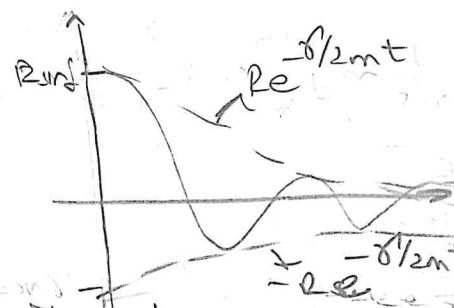
$$= \underbrace{-\frac{\gamma}{2m}}_{\text{Re}(\lambda_1)} \pm \underbrace{\frac{\sqrt{4km - \gamma^2}}{2m}}_{\text{Im}(\lambda_1)} i$$

Represents periodic motion between envelopes

1 If $A = R \cos \delta$ $B = R \sin \delta \Rightarrow R = \sqrt{A^2 + B^2}$
 $u(t) = e^{-\gamma/2m t} (R \cos \delta \cos \mu t + R \sin \delta \sin \mu t)$
 $\Rightarrow u(t) = R e^{-\gamma/2m t} \cos(\mu t - \delta)$
 $\rightarrow u(t) = R e^{-\gamma/2m t} \cos(\mu t - \delta) \rightarrow 0$ as $t \rightarrow \infty$
 $R e^{-\gamma/2m t}$, $-R e^{-\gamma/2m t}$ called envelopes
 $\tan \delta = \frac{B}{A}$



2 If $A = R \sin \delta$ $B = R \cos \delta$
 $\tan \delta = \frac{A}{B}$
 $u(t) = R e^{-\gamma/2m t} \sin(\mu t + \delta) \rightarrow 0$ as $t \rightarrow \infty$



Ex Suppose motion of spring-mass system with damping is governed by

$$u'' + \gamma u' + 25u = 0 \quad u(0) = 1, u'(0) = 0$$

Find the eqn. of motion and sketch its graph for $\gamma = 6, 10$

Solution $k = 25, m = 1, F(t) = 0$

9 $\gamma = 6$ $\Delta = \gamma^2 - 4km = 36 - 100 = -64 < 0 \Rightarrow$ underdamped motion.

$$\lambda^2 + 6\lambda + 25 = 0 \quad \lambda_{1,2} = \frac{-6 \pm 8i}{2} = -3 \pm 4i$$

$$u(t) = A e^{-3t} \cos 4t + B e^{-3t} \sin 4t$$

I.C.: $u(0) = 1 = A$

$$u'(0) = -3A + 4B = 0$$

$$\Rightarrow B = \frac{3}{4} A = 3/4$$

U → mg (*)

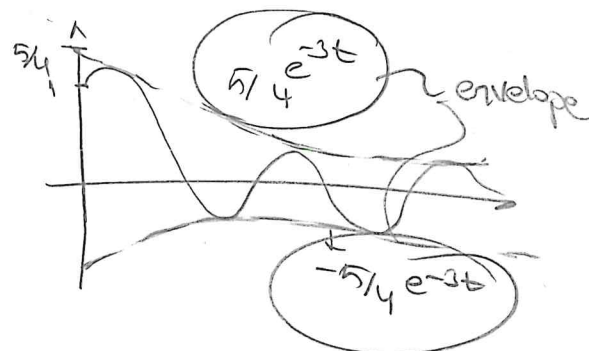
$$R = \sqrt{A^2 + B^2} = \sqrt{1 + \frac{9}{16}} = \frac{5}{4}$$

$$\tan \phi = \frac{B}{A} = \frac{3/4}{1} = \frac{3}{4} \Rightarrow \phi = \arctan \frac{3}{4}$$

$$u(t) = \frac{5}{4} e^{-3t} (\cos 4t - \phi) \quad \left. \begin{array}{l} \text{cosine} \\ \text{wave} \end{array} \right\}$$

$$R = \frac{5}{4} \quad \tan \phi = \frac{A}{B} = \frac{4}{3}$$

$$u(t) = \frac{5}{4} e^{-3t} \sin(4t + \phi) \quad \left. \begin{array}{l} \text{sine} \\ \text{wave} \end{array} \right\}$$



(b) $\gamma = 10$ $\Delta = \gamma^2 - 4km = 100 - 100 = 0$ (critically damped motion)

$$\lambda_{1,2} = -\frac{10}{2} = -5$$

$$u(t) = (A + Bt) e^{-5t}$$

$$u(0) = \underbrace{1 = A} \quad u'(0) = 0 = B e^{-5t} + (A + Bt) e^{-5t} (-5)$$

$$0 = B - 5A \Rightarrow \underline{B = 5}$$

$$u(t) = (1 + 5t) e^{-5t}$$

$$u'(t) = 5 e^{-5t} + (1 + 5t) (-5) e^{-5t}$$

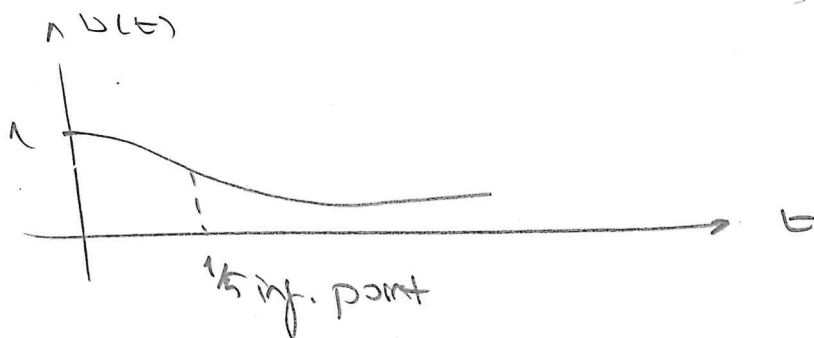
$$= e^{-5t} (5 - 5 - 25t) = -25t e^{-5t} = 0 \quad t = 0$$

$$u''(t) = -25 e^{-5t} + 125t e^{-5t}$$

$$= 25 e^{-5t} (1 + 5t) = 0$$

$$\text{if } t = 1/5$$

	$t < 0$	$t = 0$	$t > 0$	$1/5$	
u'	+	0	-	-	
u''	-	-	-	0	+



(c) $\gamma = 12$ $\gamma^2 - 4km = 144 - 100 = 44 > 0$ overdamped

$$\lambda_{1,2} = -b \pm \sqrt{11}$$

$$u(t) = A e^{(-b+\sqrt{11})t} + B e^{(-b-\sqrt{11})t} \rightarrow 0 \text{ as } t \rightarrow \infty$$

Draw picture (Exercise)



③ Forced vibration with damping

$$\gamma \neq 0 \quad F(t) \neq 0$$

$$m u'' + \gamma u' + k u = F(t) \quad (*)$$

→ Assume a periodic force $F(t) = F_0 \cos \omega t$ $F_0, \omega > 0$
 frequency of force ω
 amplitude of external force F_0

Solution of (*) has the form

$$u(t) = u_h(t) + u_p(t)$$

$\downarrow$ $\downarrow$
 homogeneous solution particular solution

$$u(t) = C_1 u_1(t) + C_2 u_2(t) + a \cos \omega t + b \sin \omega t$$

form of particular solution by using UCFM

Unless $u_1(t)$ and $u_2(t)$ are not constant $\cos \omega t$ and $\sin \omega t$

→ Assume underdamped motion $\Delta = \gamma^2 - 4km < 0$, in this case

$$u_h(t) = R e^{-\frac{\gamma}{2m} t} \cos \left(\frac{\sqrt{4mk - \gamma^2}}{2m} t - \phi \right) \rightarrow 0 \text{ as } t \rightarrow \infty$$

transient solution

$$R = \sqrt{A^2 + B^2}$$

$$\tan \phi = \frac{B}{A}$$

We can obtain $u_p(t) = a \cos \omega t + b \sin \omega t$ by substituting in (*)

as follows

$u_p(t) = a \cos \omega t + b \sin \omega t$ by substituting in (1) as follows

$$u_p(t) = \frac{F_0}{\sqrt{(k - m\omega^2)^2 + \gamma^2 \omega^2}} \sin(\omega t + \theta) \quad \text{where } \tan \theta = \frac{k - m\omega^2}{\gamma \omega}$$

Then $u_{\text{part}}(t) = \underbrace{e^{-\gamma t/2m} \cos\left(\frac{\sqrt{4mk - \gamma^2}}{2m} t - \phi\right)}_{\rightarrow 0 \quad t \rightarrow \infty} + \frac{F_0}{\sqrt{(k - m\omega^2)^2 + \gamma^2 \omega^2}} \sin(\omega t + \theta)$

transient sol. Steady state sol.

RESONANCE

→ Consider a spring-mass system without damping $\gamma = 0$

$$m u'' + k u = F(t)$$

where $F(t) = F_0 \cos \omega t$ or $F(t) = F_0 \sin \omega t$ and

$\omega_0^2 = \frac{k}{m}$ is the natural frequency of motion

$$u_{\text{general}} = u_h + u_p$$

$$u_h: \lambda^2 = -\frac{k}{m} \quad \lambda_{1,2} = \pm \sqrt{\frac{k}{m}} i = \pm i \omega_0$$

$$u_h = A \cos(\omega_0 t) + B \sin(\omega_0 t)$$

$$u_h = R \cos(\omega_0 t - \phi)$$

$$R = \sqrt{A^2 + B^2} \quad \tan \phi = \frac{B}{A}$$

OR $u_h = R \sin(\omega_0 t + \phi)$

(*) $R = \sqrt{A^2 + B^2} \quad \tan \phi = \frac{A}{B}$

We have

$u_p(t)$ if $F = F_0 \cos(\omega t)$

$$u_p(t) = t a \cos \omega t + t b \sin \omega t$$

which leads to particular solution

$$u_p(t) = \frac{F_0}{2m\omega_0} t \sin(\omega_0 t)$$

(.)

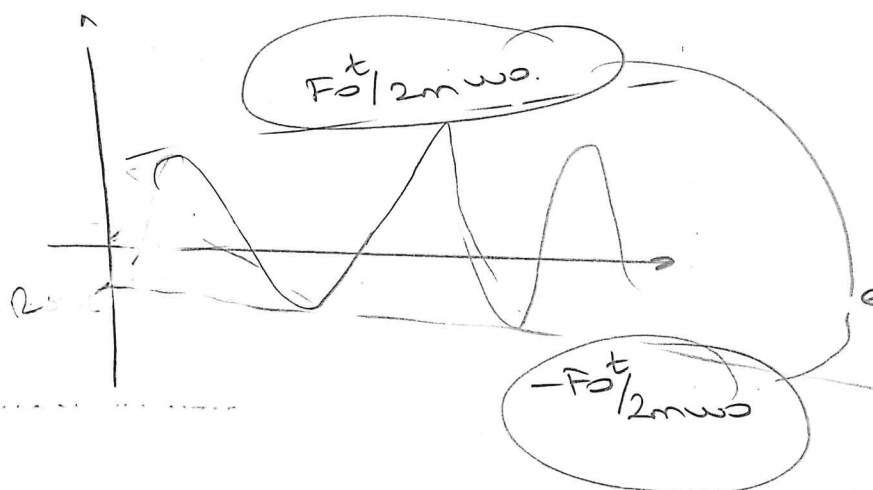
$$u_{\text{general}}(t) = \underbrace{R \sin(\omega_0 t + \phi)}_{\text{choosing } \phi} + \underbrace{\frac{F_0}{2m\omega_0} t \sin(\omega_0 t)}_{\text{particular solution}}$$

As $t \rightarrow \infty$ $u_p(t) \rightarrow \infty$
 No limit for $u(t)$ period because $u_p(t) \rightarrow \infty$
 $2\sin(\omega_0 t + \phi)$ oscillates

This case is called resonance case, when $\omega = \omega_0$

We observe resonance

frequency of force
 ↓
 Natural freq of object motion



→ solution oscillates till infinity because $u_p(t) \rightarrow \infty$
 envelope $\frac{F_0}{2m\omega_0} t \sin(\omega_0 t + \phi)$
 No limit, oscillates

Ex 7 The motion of spring-mass system is governed by

$$u'' + \gamma u' + ku = 0 \quad u(0) = 1, u'(0) = 0$$

(a) Find all γ for which the system has underdamped motion

$$\Delta = \gamma^2 - 4km < 0 \quad \gamma > 0 \quad k=4, m=1$$

$$\gamma^2 - 4 \cdot 4 < 0 \quad \gamma^2 - 16 < 0 \Rightarrow -4 < \gamma < 4$$

But, for $0 < \gamma < 4$ we have underdamped motion.

(b) Assuming $\gamma = 4$, find the position of the object at any time. Draw a graph of $u(t)$.

$$u'' + 4u' + 4u = 0$$

$$\lambda^2 + 4\lambda + 4 = 0 \Rightarrow (\lambda + 2)^2 = 0 \quad \lambda_{1,2} = -2$$

$$u(t) = (A + Bt)e^{-2t}$$

$$u(0) = 1 = A$$

$$u'(0) = 0 = B e^{-2t} + (A + Bt)(-2)e^{-2t}$$

$$0 = B - 2A \Rightarrow B = 2A = 2$$

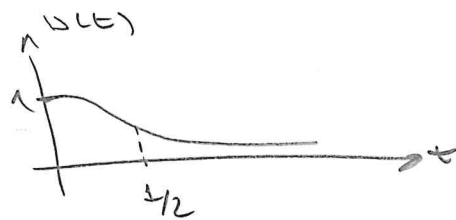
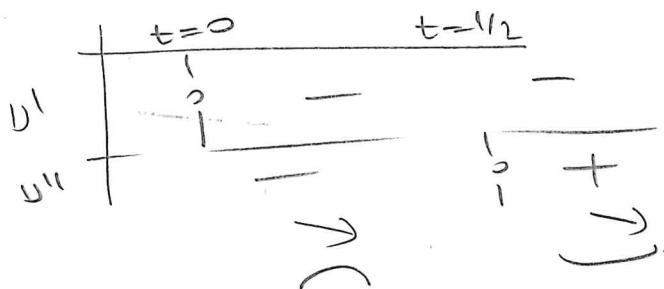
$$u(t) = (1 + 2t)e^{-2t}$$

Graph $v'(t) = 2e^{-2t} + (1+2t)(-2)e^{-2t}$

$$v'(t) = -4te^{-2t} = 0 \text{ if } t=0 \text{ - C.P.}$$

$$v''(t) = -4e^{-2t} + 8te^{-2t}$$

$$= -4e^{-2t}(1-2t) = 0 \text{ if } t=1/2 \text{ is point}$$



- (c) Assuming $\gamma=0$ (no damping) and an external force $F = 4 \sin(\omega t) + 4 \sin(3\omega t)$ find all ω for which resonance occurs (DON'T SOLVE PROBLEM)
- $$v'' + 4v = 4 \sin(\omega t) + 4 \sin(3\omega t)$$

Ch $\lambda^2 + 4 = 0$
 $\lambda = \pm 2i$

$$v_h = c_1 \cos 2t + c_2 \sin 2t$$

Resonance occurs if force frequency ω or 3ω is equal to natural frequency $\omega_0 = 2$.
 $\omega_0 = 2 = \omega$ or $\omega_0 = 2 = 3\omega$
 $\omega = 2$ or $\omega = 2/3$

we have resonance.

Ex 2 One end of a linear spring is attached to a ceiling. An object which is attached to its other end stretches the spring a distance of $1/5$ meters at equilibrium. Assume there is no damping.

- (a) write the equation of motion. You may take the gravitational acceleration $g = 9.8 \text{ m/s}^2$.

$$L = 1/5 \text{ m}, \gamma=0, m v'' + \gamma v' + k v = F(t)$$

$$mg = kL = k \Rightarrow k = \frac{mg}{L} \Rightarrow k = 5m \cdot (9.8)$$

$$m v'' + k v = 0$$

$$m v'' + 5m(9.8) v = 0 \Rightarrow \text{divide } m \Rightarrow v'' + 49v = 0$$

b) Find the general solution of equation of motion.

$$\lambda^2 + 49 = 0 \Rightarrow \lambda = \pm 7i$$

underdamped motion

$$b(t) = A \cos 7t + B \sin 7t \quad A, B \in \mathbb{R}$$

c) Suppose that the initial displacement of object is 1 m above the equilibrium position and initial velocity is 3 m/s downwards.

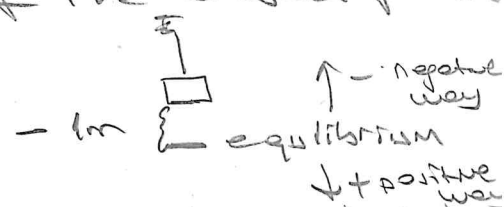
What is the max displacement of the object from equilibrium position for $t > 0$?

$$b(0) = -1 \text{ m}$$

$$b'(0) = 3 \text{ m/s}$$

$$b(0) = -1 = A$$

$$b'(0) = -7A \sin 7t + 7B \cos 7t \Big|_{t=0} = 7B = 3$$



$$R = \sqrt{A^2 + B^2} = \sqrt{(-1)^2 + \left(\frac{3}{7}\right)^2} = \sqrt{\frac{58}{49}} \text{ meters}$$

amplitude.

d) Write an example of an external forcing function which would cause resonance in the spring-mass system

$$b'' + 49b = 0$$

$$\omega_0 = \sqrt{\frac{k}{m}} = \sqrt{49} = 7$$

Resonance occurs when frequency of force

$$\omega = \omega_0$$

$F(t) = \cos 7t$ or $\sin 7t$ or any other function that contains a wave of frequency $\omega_0 = 7$.

